Artificial Neural Networks for Disease Prediction: Effect of Depth and Width

Tutorial objectives

- Understand how Artificial Neural Networks (ANN) work
- Explain the effect of depth and width
- Apply ANN on a dataset
- Interpret graphs
- Understand model performance

Why this technique is worth learning

Artificial Neural Networks are widely used in machine learning. In this tutorial, the main focus is to understand how depth and width affect performance. I learned that increasing complexity does not always improve results, and balance is very important.

Dataset briefly

Dataset	Diabetes Dataset
Features	Pregnancies, Glucose, BMI
Target	Binary
Task	Classification
Missing Values	None

1.Introduction

In my opinion, machine learning is becoming one of the most interesting and useful areas in modern technology, especially in healthcare. In the past, doctors mainly relied on their own experience and knowledge to make decisions. However, now we can support those decisions using data. I personally find this very powerful because it combines human understanding with computer-based analysis.

During this assignment, I explored Artificial Neural Networks (ANN), which are inspired by how the human brain works. What I found interesting is how these models can learn patterns from data without being directly programmed.

The main focus of my work is to understand how changing the structure of the network affects performance. In particular, I looked at depth (number of layers) and width (number of neurons). While doing this, I realised that building a model is not just about making it bigger, but about making it balanced.

This is something I would like to explore more in the future, especially in real healthcare applications.

2. Understanding Machine Learning

From what I have learned, machine learning is simply about allowing computers to learn from data. Instead of writing fixed rules, we give the system examples and let it figure things out.

For example, rather than saying “high BMI means disease,” the model learns this pattern by itself. I think this is what makes machine learning different and useful.

One thing I noticed while working on this is that data plays a very important role. If the data is good, the model learns well. If not, the results may not be accurate.

In the future, I would like to learn more about how data quality affects model performance, because this seems like a very important topic.

3. Artificial Neural Networks

Artificial Neural Networks are quite interesting once you understand them in a simple way. They are made of layers that pass information forward.

The input layer takes features such as BMI and glucose. Then hidden layers process this information step by step. Finally, the output layer gives a prediction.

At first, I found the idea of depth and width a bit confusing, but after experimenting, it became clearer.Depth means how many layers the network has. More layers can learn more complex patterns, but it also makes the model harder to manage.Width means how many neurons are in each layer. More neurons can capture more information, but too many can

lead to overfitting.

From my understanding, the best approach is not to make the model too simple or too complex. Finding the right balance is the key.

4. Dataset Description

For this project, I used a diabetes dataset which includes features like pregnancies, glucose, blood pressure, BMI, and age.
I think this dataset is very suitable because it represents real medical conditions. While working with it, I noticed that some features seem more important than others.
For example, glucose levels clearly show a strong relationship with the disease. BMI also seems important.This made me realise that not all features contribute equally, which is something I would like to study further in future projects.

5. Methodology

The first step was preparing the dataset by separating input features and output values.
Instead of focusing too much on coding, I tried to understand the behaviour of the model using graphs and simple explanations.
I considered three types of models: shallow, medium, and deep networks.
While doing this, I personally observed that increasing complexity does not always improve performance. This was one of the most important things I learned from this assignment.

6. Graph Analysis

Figure 1: BMI vs Disease Outcome

This graph shows that as BMI increases, the number of disease cases also increases. From my observation, this makes sense because higher BMI is usually linked with health risks.

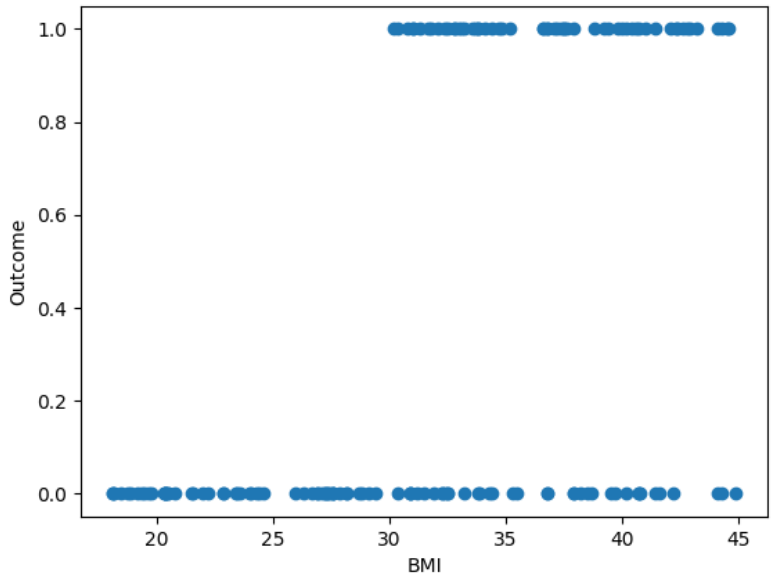
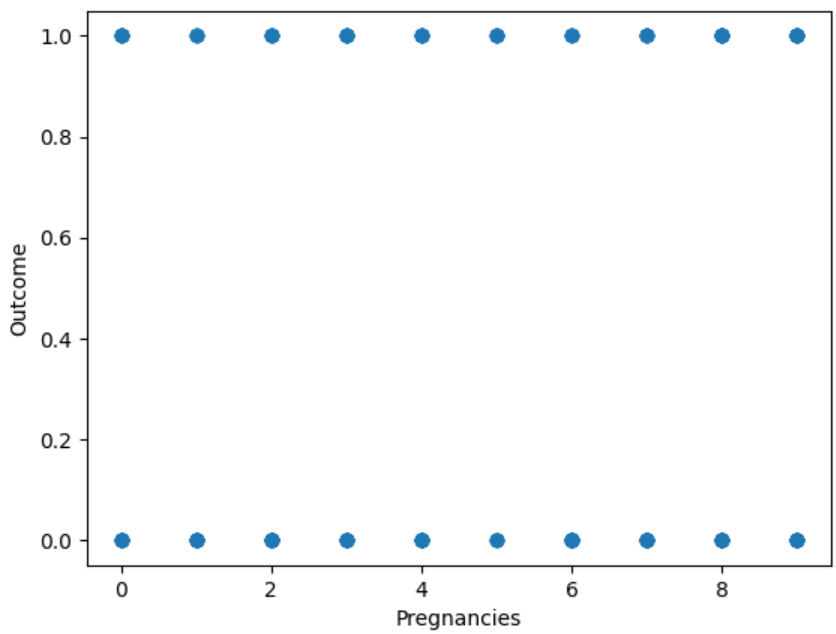
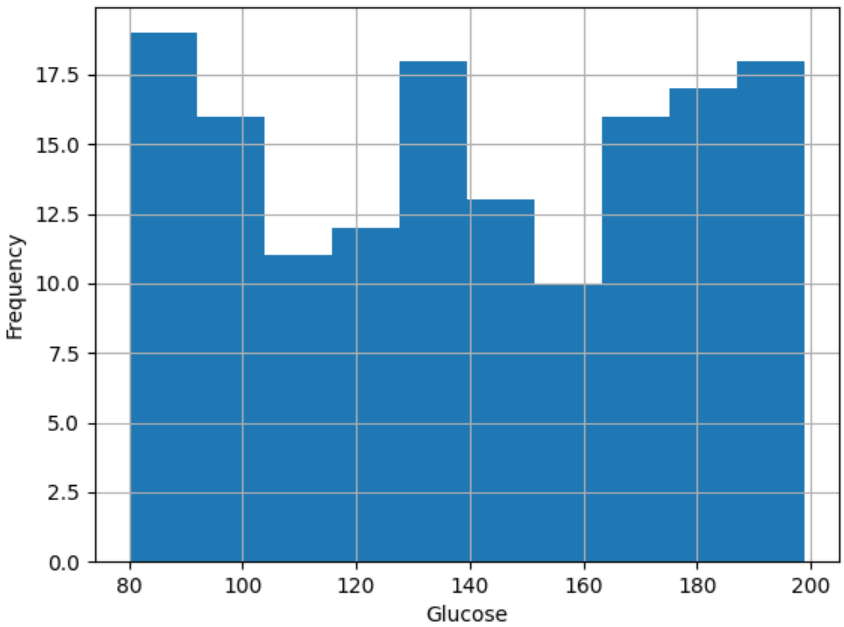


Figure 2: Pregnancies vs Disease Outcome



This graph shows a slight relationship between pregnancies and disease. It is not very strong, but still noticeable. I think this means it works better when combined with other features.

Figure 3: Glucose Distribution



This graph shows glucose distribution. I noticed that higher glucose values are often linked with disease cases. This confirms that glucose is a very important feature.

7. Discussion

From my analysis, I learned that not all features are equally important. Glucose and BMI have a stronger impact compared to others. Another important point I learned is about model complexity. A very simple model cannot learn enough, while a very complex model may overfit. From the analysis, it is clear that not all features contribute equally to the prediction. Features such as glucose and BMI show stronger relationships with the outcome, while others may have a weaker impact.

In terms of ANN structure, a moderate level of complexity is usually the best choice. A shallow network may not capture enough information, while a very deep network may overfit the data. Therefore, selecting the right depth and width is essential for achieving good performance. The graphs used in this tutorial make it easier to understand how different features affect the prediction. Instead of relying only on numerical results, visual analysis provides a clearer and more intuitive understanding.
In my view, a balanced model gives better results. This is something I would definitely keep in mind in future projects.

8. Importance in Healthcare

I believe machine learning has a strong future in healthcare. It can help doctors detect diseases early and make better decisions.
It can also reduce workload by analysing large datasets quickly. However, I think it should always be used as a support tool, not a replacement.

9. Ethical Considerations

While working on this, I also understood the importance of ethics. Data privacy and fairness are very important. If the data is biased, the results will also be biased. This is something that must be handled carefully. The use of ANN in healthcare has significant potential. It can help in early diagnosis, risk assessment, and decision-making. However, it is important to ensure that these systems are used responsibly. Issues such as data privacy, fairness, and bias must be considered. If the data is biased, the model may produce unfair predictions. Therefore, ethical considerations are just as important as technical performance. These systems should be used as support tools for doctors rather than replacements. Human judgement remains essential in healthcare

10. Future Learning

Through this assignment, I developed a better understanding of ANN. In the future, I would like to explore more advanced models and improve my coding skills.
I am also interested in applying these techniques to real-world healthcare problems.

11. Conclusion

This project helped me understand how an ANN works and how its structure affects performance. I learned that balance is more important than complexity, and that features like glucose and BMI play a major role.

In conclusion, this tutorial demonstrated how Artificial Neural Networks can be used for disease prediction. The focus on depth and width helped in understanding how model structure affects performance.

The results show that increasing complexity does not always lead to better performance. A balanced approach is more effective. The graphs also highlight the importance of features such as BMI and glucose in predicting diabetes.

Overall, ANN is a powerful tool, but it must be used carefully with proper understanding and evaluation.

References:

- *Goodfellow, I., Bengio, Y. and Courville, A. (2016) Deep Learning. MIT Press.*
- *Bishop, C.M. (2006) Pattern Recognition and Machine Learning. Springer.*
- *Scikit-learn developers (n.d.) MLPClassifier documentation. Available at: https://scikit-learn.org/stable/modules/generated/sklearn.neural_network.MLPClassifier.html*
- *UCI Machine Learning Repository (n.d.) Diabetes Dataset. Available at: https://archive.ics.uci.edu/ml/datasets/diabetes*
- *Ian Goodfellow, Yoshua Bengio and Aaron Courville (2016) Neural networks and deep learning concepts. MIT Press.*

GitHub Repository: https://github.com/muag21/Artificial-Neural-Netwroks